

|               |                                                                |     |    |     |
|---------------|----------------------------------------------------------------|-----|----|-----|
| <b>11 (a)</b> | <u>pulse</u> of ultrasound                                     | (1) |    |     |
|               | reflected at boundaries / boundary                             | (1) |    |     |
|               | received / detected (at surface) by transducer                 | (1) |    |     |
|               | signal processed and displayed                                 | (1) |    |     |
|               | time between transmission and receipt of pulse gives           |     |    |     |
|               | (information about) depth of boundary                          | (1) |    |     |
|               | reflected intensity gives information as to nature of boundary | (1) |    |     |
|               | <i>(any four points, 1 each, max 4)</i>                        |     | B4 | [4] |

|                |             |                                        |    |     |
|----------------|-------------|----------------------------------------|----|-----|
| <b>(b) (i)</b> | coefficient | $= (Z_2 - Z_1)^2 / (Z_2 + Z_1)^2$      | C1 |     |
|                |             | $= (6.3 - 1.7)^2 / (6.3 + 1.7)^2$      | A1 |     |
|                |             | $= 0.33$ <i>(unit quoted, then -1)</i> |    | [2] |

|             |          |                                         |    |     |
|-------------|----------|-----------------------------------------|----|-----|
| <b>(ii)</b> | fraction | $= \exp(-\mu x)$                        | C1 |     |
|             |          | $= \exp(-23 \times 4.1 \times 10^{-2})$ |    |     |
|             |          | $= 0.39$                                | A1 | [2] |

|              |           |                                 |    |     |
|--------------|-----------|---------------------------------|----|-----|
| <b>(iii)</b> | intensity | $= 0.33 \times 0.39^2 \times I$ | C1 |     |
|              |           | $= 0.050 I$                     | A1 | [2] |

*(do not allow e.c.f. from (i) and (ii) if these answers are greater than 1)*

|               |                                                                         |  |    |     |
|---------------|-------------------------------------------------------------------------|--|----|-----|
| <b>12 (a)</b> | loss / reduction in power / energy / voltage/ amplitude (of the signal) |  | B1 | [1] |
|---------------|-------------------------------------------------------------------------|--|----|-----|